

1	Course Name:
	Advanced Machine Learning
2	Course Code:
	IRG61104
3	Year and Semester Offered:
	Year 1 Semester 2
4	Credit Hours Value:
	4 Credit Hours [4 SLT Credit]
5	Prerequisite(s), if applicable:
	NA
6	Course Synopsis:
	This course is related to the study of optimization, and computer science to create automated systems to make predictions or decisions without human intervention. Topics covered include data analysis supervised and unsupervised learning, artificial neural networks, genetic algorithm and real world machine learning application.
7	Course Learning Outcome:
	Upon completion of the course, students should be able to: 1. Determine the pattern produced. 2. Performs data attributes analysis using machine learning tools. 3. Practices an ability to discuss the patterns in decision making.
8	Content Outline:
	Topic
	1. Machine learning Overview
	2. Linear methods for regression
	3. Artificial Neural Networks
	4. Genetic Algorithm
	5. Kernel functions and support vector machines)
	6. Unsupervised machine learning
	7. Survey of advanced topics
9	References:
	1. Yuxi (Hayden) Liu; Phyton Machine Learning by Example; Packt Publishing, 2017. 2. Rogers Simon, Girolami, Mark; A First Course in Machine Learning; CRC Press, 2016, ISBN 9781498738484